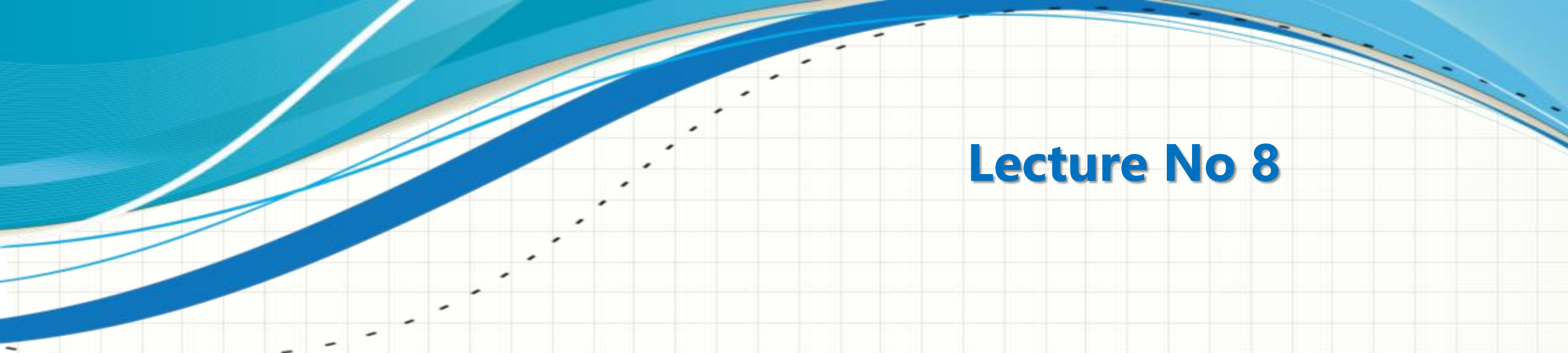




CSC432 – INFORMATION SECURITY

Dr. Muhammad Sharjeel

muhammadsharjeel@cuilahore.edu.pk



Lecture No 8

ASYMMETRIC KEY CRYPTOGRAPHY AND RSA

Issues with Symmetric Key Cryptography



- ❑ The problem with symmetric cryptography was the 'key transfer'
- ❑ The dilemma: In order to transfer a secret, a different secret (the key) had to be delivered before
 - ❑ How can the key be sent from one party to another?
 - ❑ If you sent it over an insecure channel (i.e. telephone, internet, etc.) what's to stop someone from intercepting and using it?
- ❑ Diffie-Hellman showed a clever manner how to overcome the dilemma

Diffie-Hellman Key Exchange



- ❑ Two mathematicians Whitfield Diffie and Martin Hellman, in 1976, devised the notion of Public Key Cryptography in "New Directions in Cryptography"
- ❑ They showed how two communication partners that are at different locations can publicly create a common key without fearing that a third person would crack it
- ❑ This was the first major step in Public Key Cryptography

Diffie-Hellman Key Exchange



1. Alice and Bob publicly pick 2 integers: a prime number **p** and an integer **s** between 1 and p
2. Alice picks a random number **a** that is less than **p**
2. Bob picks a random number **b** that is less than **p**
3. Alice computes **A = s^a MOD p** and sends A to Bob
3. Bob computes **B = s^b MOD p** and sends B to Alice
4. Alice computes the key **K = B^a MOD p**
4. Bob computes the key **K = A^b MOD p**

Diffie-Hellman Key Exchange – Example



1. Alice and Bob publicly pick 2 integers: $p = 11$ and $s = 3$

2. Alice picks $a = 2$

2. Bob picks $b = 4$

3. Alice computes
 $A = 3^2 \text{ MOD } 11 = 9$

3. Bob computes
 $B = 3^4 \text{ MOD } 11 = 4$

4. Alice computes the key $K = 4^2 \text{ MOD } 11 = 5$

4. Bob computes the key $K = 9^4 \text{ MOD } 11 = 5$

Diffie-Hellman Key Exchange



We have the following two questions:

- ☐ Why do Alice and Bob always end up with the same key K ?
- ☐ Why can't an eavesdropper compute K ?

Diffie-Hellman Key Exchange



Why do Alice and Bob always end up with the same key K?

Alice and Bob compute the key K in the final step as follows:

$$\text{Alice: } K = B^a \text{ MOD } p = (s^b \text{ MOD } p)^a \text{ MOD } p = s^{ba} \text{ MOD } p$$

$$\text{Bob: } K = A^b \text{ MOD } p = (s^a \text{ MOD } p)^b \text{ MOD } p = s^{ab} \text{ MOD } p$$

□ Since $s^{ba} = s^{ab} \text{ MOD } p$ both Alice and Bob compute the same key K

Diffie-Hellman Key Exchange



Why can't an eavesdropper compute K?

- ❑ In order to uncover the key K an eavesdropper has to compute either a or b
- ❑ Mathematics teaches that this is impossible, if a, b, p, s, A, B were chosen as of **1024-bits** numbers or larger
- ❑ The reason: although it is quite simple to compute i.e. $A = s^a \text{ MOD } p$, however, solving this equation for a is impossible
- ❑ More formally stated: Although the "discrete exponential function" can be executed, its inverse, the "discrete logarithm function" can not
- ❑ Hence, the "discrete exponential function" is an example of a "One-Way function"

Public Key Cryptography



- ❑ Greatest and perhaps the only true revolution in the entire history of cryptography
- ❑ From its beginnings to modern times, virtually all cryptography have been based on the elementary tools of substitution and permutation, and can be classed as private/secret/single key (symmetric) cryptography
- ❑ All classical, and modern block and stream ciphers are of this form

Birth of Public-Key Cryptosystems



- ☐ 1960's: NSA secretly discovered public-key cryptography
- ☐ 1970: First known (secret) report on public-key cryptography by CESG, UK
- ☐ 1976: Diffie and Hellman public introduction to public-key cryptography
- ☐ Avoid reliance on third-parties for key distribution
- ☐ Allow digital signatures

Principles of Public-Key Cryptography



- ❑ Symmetric algorithms used same secret key for encryption and decryption
- ❑ Asymmetric algorithms in public-key cryptography use one key for encryption and different but related key for decryption
- ❑ Characteristics of asymmetric algorithms:
 - ❑ Computationally infeasible to determine decryption key given only algorithm and encryption key
 - ❑ Either of two related keys can be used for encryption, with other used for decryption

Principles of Public-Key Cryptography



- ❑ Use of two keys has profound consequences in the areas of confidentiality, key distribution, and authentication
- ❑ Works by the clever use of number theory problems that are easy one way but hard the other
- ❑ Note that public key schemes are;
 - ❑ Not more nor less secure than private key (security depends on the key size for both)
 - ❑ They do not replace private key schemes (they are too slow to do so)
 - ❑ Both complement each other



RSA

RSA



□ Given sufficiently large key lengths, RSA realizes the

"The knowledge of the encoding key does not reveal the knowledge of the decoding key. Even the usage of the most powerful computers combined will not suffice to crack the secret decoding key based on the knowledge of the publicly known encoding key."

RSA Cipher



- ❑ Three MIT researchers **R**ivest, **S**hamir and **A**dleman came up with their idea for the RSA Ciphers in 1977
- ❑ Best known (unbreakable cipher) public key system widely used in electronic commerce protocols, and is believed to be secure given sufficiently long keys and the use of up-to-date implementations
- ❑ RSA Cipher is quite easy to understand, don't be mislead, the fact that a cipher can be executed easily does not make it insecure

RSA Cipher



RSA Encryption and Decryption

Step 1: Preparation

- ❑ Choose two (large and distinct) primes **p** and **q** so that their product **$n=p*q$** is greater than the used alphabet length M (i.e. here M=26)
- ❑ Compute **$\phi(n)$**

$$\phi(n) = (p-1)(q-1)$$

RSA Cipher



RSA Encryption and Decryption

Step 2: Encryption uses the public key (e, n)

- ❑ Choose a public encoding key e (such that $1 < e < \phi(n)$) relatively prime to $\phi(n)$
- ❑ We now encrypt each plain letter P by computing

$$C = P^e \text{ MOD } n$$

RSA Cipher



RSA Encryption and Decryption

Step 3: Decryption uses the private key (d, n)

- ❑ The private decoding key **d** is chosen as the inverse of **e MOD $\phi(n)$**

$$(e * d) \text{ MOD } \phi(n) = 1$$

- ❑ In other words, $ed-1$ is exactly divisible by $\phi(n)$

- ❑ We decrypt by computing

$$P = C^d \text{ MOD } n$$

RSA Cipher



RSA Encryption and Decryption - Example

- Lets see how the word "SAFE" is encrypted/decrypted using the RSA Cipher

Step 1: Preparation

- Choose two (large and distinct) primes **p** and **q** so that their product **n=p*q** is greater than the used alphabet length M (i.e. here M=26)
- Say **p=3** and **q=11**, then **n=33**
- Compute $\phi(n)$

$$\phi(n) = (p-1)(q-1)$$

$$\phi(33) = (3-1)*(11-1) = 20$$

RSA Cipher



RSA Encryption and Decryption - Example

Step 2: Encryption uses the public key (e, n)

❑ Choose a public encoding key e (such that $1 < e < \phi(n)$) relative prime to $\phi(n)$

❑ Here, possible values for e are 3, 7, 9, 11, 13, 17, 19, let's pick $e=3$

❑ We now encrypt each plain letter P by computing

$$C = P^e \text{ MOD } n$$

$$S=18: 18^3 \text{ MOD } 33 = \mathbf{24}$$

$$A=0: 0^3 \text{ MOD } 33 = \mathbf{0}$$

$$F=5: 5^3 \text{ MOD } 33 = \mathbf{26}$$

$$E=4: 4^3 \text{ MOD } 33 = \mathbf{31}$$

RSA Cipher



RSA Encryption and Decryption - Example

Step 3: Decryption uses the private key (d, n)

- ❑ The private decoding key d is chosen as the inverse of $e \text{ MOD } \phi(n)$

$$(e * d) \text{ MOD } \phi(n) = 1$$

- ❑ In other words, $ed-1$ is exactly divisible by $\phi(n)$
- ❑ since $(3*7) \text{ MOD } 20 = 1$, so $d = 7$

- ❑ We decrypt by computing

$$P = C^d \text{ MOD } n$$

$$24^7 \text{ MOD } 33 = 18, \quad 18=S$$

$$26^7 \text{ MOD } 33 = 5, \quad 5=F$$

$$0^7 \text{ MOD } 33 = 0, \quad 0=A$$

$$31^7 \text{ MOD } 33 = 4, \quad 4=E$$

RSA Cipher – A Secure Cipher



- ❑ RSA (actually) does not fulfill the public key property
 - ❑ the decoding key can be derived from the encoding key and $\phi(n)$, if p and q are chosen to be small numbers
- ❑ But RSA encryption system can be upgraded so that even though everybody knows the modulus n , not even the best eavesdropper is able to compute $\phi(n)$.
- ❑ Solution:
 - Choose the two primes, p and q to be at least **1024-bit numbers**

RSA Cipher – A Secure Cipher



- ☐ Choosing the key length wisely becomes crucial when transferring sensitive information using RSA
- ☐ A public key length of 768 bits (232 decimal) or 1024 bits (309 decimal) digits is recommended today
- ☐ The payoff here is speed
- ☐ Greater key sizes slow the computerized encryption process down

RSA Cipher – A Secure Cipher



- ❑ No eavesdropper (not even the NSA or the FBI) is able to compute $\phi(n)$ from the publicly known modulus n since its factors p and q (required to compute $\phi(n)$ as $(p*q) = (p-1) * (q-1)$) can not be derived from n
- ❑ Rivest, Shamir and Adleman (experienced computer experts) knew that;

“multiplication of two large numbers is not difficult, however, finding the factors of a given large integer is a difficult computer problem”

RSA Cipher – A Secure Cipher



- ❑ Among the many possible ways of attacking the RSA Cipher, factoring is the most promising one
- ❑ But the fact is that sufficiently large integers can not be factored, even when using the best factoring algorithms on the fastest computers in the world
- ❑ Consequently, if n is chosen sufficiently large nobody is able to find the factors p and q of n and thus nobody can compute $\phi(n)$ in order to then identify the decoding key d
- ❑ **Since finding the factors of a huge integer n [without knowing $\phi(n)$] is difficult, the RSA encryption is secure**

RSA Cipher – A Secure Cipher



- ❑ The inability to crack RSA is based mathematically on the inability to factor the product n into the two huge primes p and q
- ❑ However, the reverse process, computing the product given the two primes is easy
- ❑ A function with such property is called a one-way function
- ❑ A function that computes the result (i.e. the product of two primes) easily. However, given the result it is practically impossible to compute the original values
- ❑ Mathematically stated: Given x , $f(x)$ can be computed easily. However, given $f(x)$, computing x is practically impossible

RSA Cipher – A Secure Cipher



- ❑ The status quo is: Cracking the RSA encryption is at most as difficult as factoring huge numbers
- ❑ Albrecht Beutelspacher, a German cryptographer, thinks that “Introducing a crypto-system whose strength is based on the fact that no Mathematician has come up yet with a reasonable factoring algorithm is a massive insult for the Mathematicians.”

Symmetric VS Asymmetric Key Cryptography



Which method provides stronger security?

- ☐ With suitable keys and algorithms, both methods can be secure enough for most purposes

Which method is more convenient?

- ☐ To use symmetric cryptography, both parties must know the secret key, which can be quite inconvenient
- ☐ To use asymmetric key cryptography, one only needs to find the public key to communicate with someone else, which can be a lot more convenient

Which method performs better?

- ☐ Encrypting and decrypting a lot of information with asymmetric key cryptography can be painfully slow in comparison to symmetric cryptography

Hybrid Cryptography



- ❑ Ideally, we would like to combine the strengths of symmetric and asymmetric key cryptography, and avoid their weaknesses
- ❑ We want the efficiency of symmetric cryptography combined with the ease of use and convenience of asymmetric key cryptography
- ❑ When two parties want to communicate securely, asymmetric key cryptography is used to exchange a random symmetric session key
- ❑ Since the session key is encrypted, we can ensure secrecy and mutual authentication
- ❑ Thus, key distribution and setup is easy and risk-free

Hybrid Cryptography



- ❑ To communicate, symmetric cryptography is used with the session key
 - ❑ Since only these two parties know the session key, the messages are secure
 - ❑ Since secret key cryptography is used, this can be done relatively efficiently
- ❑ When done, both parties destroy the session key. If communication is required in the future, this process is repeated from the beginning to obtain a completely new session key
- ❑ This general approach is used in **Pretty Good Privacy (PGP)** and elsewhere
- ❑ The fast encryption speed of symmetric ciphers is coupled with the security of asymmetric ciphers

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THANKS